Melvin Khakabo

Software Engineer | AI Engineer
+ (254) 740-155-060 · melvinkhakabo@gmail.com · https://www.linkedin.com/in/melvin-khakabo/
Nairobi, Kenya

PROFESSIONAL SUMMARY

Software and AI engineer building practical learning technology for African contexts. Experienced across full-stack development (Python, Django, Node.js, React) and LLM-powered systems, including multi-agent workflows, prompt engineering and AI evaluation. Years of teaching first-time coders and corporate teams inform how I design tools that real learners and educators can use.

EXPERIENCE

AI Engineer & Founding STEM/AI Instructor **Feb 2025 - Present**
Learning Sprouts (now The Learning Quarter), Nairobi

- Design and build LLM-powered tools and multi-agent workflows for education, including Ujuzi Agent, a lesson-plan generation system for African classrooms.
- Integrate LLMs into teaching workflows through prompt engineering, agent orchestration and evaluation of output quality and reliability.
- Launched 5+ AI and coding course streams for young learners and corporate clients in Nairobi and Lagos, and developed 15+ project-based curricula.
- Lead a team of 10+ instructors and run large-scale programmes, including AI Hackathon and PUMaC Africa (the official Africa edition of Princeton University's Mathematics Competition), serving 500+ participants.
- Secured AI training contracts with 5+ Kenyan corporations by translating technical value for non-technical decision-makers.

Data Scientist **August 2023 - May 2024**
HackBio

- Analysed and visualised complex cancer datasets in R and Python, building and maintaining reproducible data-processing scripts.
- Built supervised (Random Forest) and unsupervised (K-Nearest Neighbors) models across the full lifecycle (training, testing, validation and interpretation), achieving 95% prediction accuracy.
- Coordinated 100+ development tasks and 8+ project timelines in Jira and Trello, delivering 90% of milestones on time.

Web Accessibility Developer Contractor **January 2024 - February 2024**
Technoprise Global

- Implemented WCAG 2.1-compliant accessibility features in ReactJS applications for users with disabilities.
- Built reusable custom React components with ARIA roles and properties to improve screen-reader compatibility.

Technical Support Agent **August 2022 - July 2023**
ALX

- Supported learners through phone calls, WhatsApp, live chat and direct calls during application and onboarding, contributing to a 100% assessment completion rate.
- Troubleshot learning management system and portal issues, resolving IT tickets within 12 hours on average.

Full Stack Developer **July 2021 - August 2022**
TheJitu Ltd

- Developed scalable back ends in Python Django and Node.js (TypeScript) with SQL Server and PostgreSQL for high-traffic applications.
- Built APIs using multi-threading and RabbitMQ queues for email and SMS notifications.
- Developed responsive front ends in React and Angular, integrated with Django and Node.js services.

PROJECTS

Ujuzi Agent | *AI Engineer*

- Multi-agent AI system, built on Google's Agent Development Kit, that generates localised lesson plans for African classrooms, aligned to local curricula, languages and student needs.
- Designed agent orchestration, prompt engineering and LLM integration, with evaluation methods to check

output quality and curriculum fit.

Learning Logs | *Full-Stack Developer*

- React web app where learners log topics and journal their progress, with cloud-backed storage and automated topic recommendations based on predefined criteria.

User Management System | *Full-Stack Developer*

- Project and task management application with Node.js and Express APIs and a React front end, supporting project creation, user assignment and task allocation.

EDUCATION

BSc Biomedical Science and Technology
Egerton University · Nakuru, Kenya · 2023

CERTIFICATIONS

- **Cybersecurity** - ALX · 2025
- **AWS Cloud Computing** - AWS · 2025
- **Software Engineering** - ALX, Power Learn Project · 2022, 2024
- **Data Science and Machine Learning** - HackBio · 2024
- **Aspire Leadership Program** - Harvard · 2023
- **AI Career Essentials + AI Toolkit** - ALX Africa · 2023
- **Agile Project Management** - Alison · CPD Certified · 2023
- **Computer Hardware Basics** - Cisco · 2023

VOLUNTEER EXPERIENCE

Volunteer Technical Tutor **July 2021 - Present**

- Mentor students in programming and software development practices through code reviews, debugging support, organized hackathons, and co-designed learning materials.

SKILLS

- **Languages:** Python, R, JavaScript, TypeScript, SQL
- **AI/ML:** LLM integration, multi-agent systems, Google Agent Development Kit (ADK), prompt engineering, AI evaluation, machine learning (Random Forest, KNN), data analysis and visualisation
- **Web & Back end:** Django, Node.js, Express, React, Angular, REST and GraphQL APIs, RabbitMQ
- **Databases & DevOps:** PostgreSQL, SQL Server, MySQL, SQLite, Git, CI/CD (GitHub Actions, Bitbucket Pipelines), AWS
- **Accessibility:** WCAG 2.1, ARIA
- **Collaboration:** Jira, Trello, Slack, Microsoft Teams, Agile, technical training and curriculum design